

EXAMPLES – SAMPLING THEORY AND CONFIDENCE INTERVAL

1. It is asked for the estimation of the lamps' defection proportion. So that the difference between the exact and estimated proportion is not more than 0.03, determine the sample size for the 95% confidence interval
- a) If we know from past experiences that the sample proportion is 0.05.
 - b) If we have no idea about sample proportion.

Ans: a) $n = 203$ b) $n = 1068$

2. The diameter of the screws produced in a factory has a normal distribution with a standard deviation of 0.38 cm. To estimate the mean of screw diameters, 12 random screws are selected, and the 95% confidence interval is computed as

$$1.64 \leq \mu \leq 2.36$$

However, the factory manager asks for a more definite interval. So that the difference between the exact value and the estimated value is not more than 0.08 cm, determine the sample size for

- a) 95% confidence interval.
- b) 99% confidence interval.

Answer: a) $n=87$ b) $n=150$

3. X_1 and X_2 represent the elements of two samples that are randomly selected from a population whose mean is μ and variance is σ^2 . Four estimators of μ are given below:

$$\hat{\mu}_1 = \frac{1}{2}X_1 + \frac{1}{2}X_2, \quad \hat{\mu}_2 = \frac{2}{5}X_1 + \frac{3}{5}X_2, \quad \hat{\mu}_3 = \frac{1}{5}X_1 + \frac{2}{5}X_2, \quad \hat{\mu}_4 = \frac{1}{4}X_1 + \frac{3}{4}X_2$$

- a) Find the unbiased estimators.
- b) Which estimator is the most effective?
- c) Compute the relative efficiency of $\hat{\mu}_1$ with respect to $\hat{\mu}_2$ and interpret the results.

Answer: a) $\hat{\mu}_1, \hat{\mu}_2, \hat{\mu}_4$ b) $\hat{\mu}_1$ c) 1.04, if we use $\hat{\mu}_2$ instead of $\hat{\mu}_1$, 1.04 times larger sample size we need.

4. An operator's long-distance phone calls are normally distributed with a mean of 8 minutes and a standard deviation of 2 minutes. If you choose samples of 25 calls, what percentage of these sample means will be between 7.8 and 8.2 minutes?

Ans: 38.30%

5. An n -unit sample will be taken from a population that is normally distributed with a standard deviation of 8. What n should be so that the difference between the sample mean and the population mean is at most 1.5 with a probability of 0.95?

Answer: $n=109$

6. A sample of size 400 units is selected without replacement from a population of size 5000 units. The standard deviation of the population is $\sigma = 50$. The sample mean is calculated as 4500. Estimate the population mean with 0.90 confidence level.

Answer: [4496.105, 4503.895]

7. The weights of 10000 products in a warehouse are examined. A sample of 36 units of these products was selected (without replacement), the mean of which was 20 gr. and its standard deviation is 4 gr. found. Estimate the weight mean of the products in the warehouse with confidence level 0.70.

Answer: [19.307, 20.693]



8. If $N=5000$, $n=49$, $\sum_{i=1}^{49} X_i = 2940$ and $\alpha=0.05$, then determine confidence interval for μ . ($\sigma = 7$)
Answer: [58.04, 61.96]
9. The proportion of defective products produced by machine A is investigated. For this purpose, 180 out of 2000 products produced by the machine were chosen randomly, and it was understood that 27 of them were faulty. Estimate the proportion of defective products produced by machine A with a 99% confidence interval (selection is made without replacement).
Answer: [0.085, 0.21]
10. A sample of size 12 is selected from a population distributed normally. If the standard deviation of the sample is 16.8, then construct the 90% confidence interval for the population variance
Ans: [157.8, 679.4]
11. According to a survey, 72 of 225 people will vote for the political party A. Construct a 99% confidence interval for the proportion of all voters that will vote for party A.
Ans: [0.24, 0.40]
12. To compare the average life of two tires, 55 tires are selected in brand A and 48 tires are selected in brand B. The mean and standard deviation of the first sample are 20000 km and 950 km, respectively, as well as the mean and standard deviation of the second sample are 17000 km and 1000 km, respectively. If the lifetimes of tires are distributed normally, construct a 95% confidence interval for the difference of means of the two brands' lives. Can we claim that there is no statistically significant difference between means?
Ans: [2621.75, 3378.25], there is an significant difference.
13. In a school, 60 of 75 female students and 55 of 80 male students pass the English exam. Construct a 99% confidence interval for the difference in population proportion. Can we claim that there is no statistically significant difference between female and male students with regard to their success in English exams?
Ans: [-0.07, 0.29], there is no significant difference.
14. The diameter of the screws produced in a factory has a normal distribution. To estimate the mean of screw diameters, 8 random screws whose diameters are 7, 11, 11, 9, 10, 13, 12 and 10 mm are selected. Construct a 95% confidence interval for the mean diameters of all screws produced in this factory if
a) The population variance is 3 mm.
b) The population variance is unknown.
c) Solve the same problem for a 92% confidence interval (for the unknown population variance)
Answer: a) $9.175 \leq \mu \leq 11.575$ b) $8.831 \leq \mu \leq 11.919$ c) $9.015 \leq \mu \leq 11.735$
15. To estimate the monthly rent amount of flats in a region of Istanbul, a sample of 35 units is randomly selected. If it is known that the standard deviation of monthly rent amounts is 28 million TL,
a) Find the probability that the difference between the sample mean and population mean is at most 8 million TL.
b) Find the sample size so that the probability calculated in (a) would be 0.99.
Answer: a) 0.9090 b) $n=82$



16. Let us assume that 25% of people in a country are against the construction of nuclear power plants. If a sample of 40 people is randomly selected, find the probability that the sample proportion of people that against to the construction of nuclear power plant is more than 0.27?

Answer: 0.3859

17. It is claim that the variance of a normally distributed population is 32. A sample of 16 units is selected from the population. If the sample variance is smaller than 13.355 or larger than 65.280, this claim is rejected. If population variance is really 32, find the probability that the claim is rejected.

Answer: 0.035

18. The mean and standard deviation of weights of 100 products manufactured in a factory were measured as 1040 gr and 25 gr, respectively. Determine the 95% confidence interval for the mean weights of the products produced in this manufacturing process.

Ans: [1035.1, 1044.9]

19. The mean of a random sample of size $n = 25$ is 50. Given that the standard deviation of the normally distributed population is 10, construct the 95% confidence interval for the population mean.

Ans: [46.08, 53.92]

20. A light bulb company produces a new type of light bulb. 100 bulbs are randomly selected from the production lines and their mean and standard deviation are measured as 1280 hours and 140 hours, respectively. Construct the confidence interval for the population mean for $\alpha = 0.05$.

Ans: [1252.56, 1307.44]

21. In a sample of 400 high school students, 32 students passed the university entrance exam. Find the 95% confidence interval for the university students' success rate.

Ans: [0.053, 0.107]

22. In a foreign language course, computer-aided education is given in class A, and education with classical methods is given in class B. Six weeks after the beginning of the course, the same exam was applied to both classes and the results were compared. The mean and standard deviation of the grades of 40 students randomly selected from Class A is 86 and 12 while the mean and standard deviation of the grades of 35 students randomly selected from Class B is 72 and 14. Determine the 99% confidence interval of the difference between the mean grades of the students in both classes.

Ans: [6.18, 21.82]

23. It is desired to check whether the proportions of two different drugs curing a disease are different. For this purpose, drugs A and B were tested on 1000 patients each. It was observed that 825 and 760 of the patients who were administered A and B drugs, respectively, recovered at the end of the treatment. Accordingly, determine the 95% confidence interval for the difference in the proportions of the drugs to cure the disease.

Ans: [0.029, 0.10]

24. A sample of 16 units was selected from a population known to be normally distributed, and the sample mean and standard deviation were calculated as 35 and 4.6, respectively. Estimate the population mean with a confidence 0.95.

Ans: [32.55, 37.45]

25. A new product is offered for sale in 8 regions whose populations are close to each other. In these regions, the monthly sales mean of the product is calculated as 20500 and the standard deviation as 1600. Estimate the population variance and standard deviation with a confidence level of 0.90.

Ans: $1273903.46 \leq \sigma^2 \leq 8269497$ and $1128.67 \leq \sigma \leq 2875.67$

26. The mean and standard deviation of 10 goods with brand A are calculated as 1.5 and 0.4 while the mean and standard deviation of 11 goods with brand B are calculated as 2.0 and 0.6. Estimate the ratios of population variances and standard deviations with a confidence level of 0.98.

Answer: $0.09 \leq \sigma_1^2 / \sigma_2^2 \leq 2.34$ and $0.30 \leq \sigma_1 / \sigma_2 \leq 1.53$

27. The proportion of families using property A in a region is investigated. It is estimated that the proportion of families using property A to total families in the region is 0.40. Determine the number of families to be sampled so that the difference between the sample proportion and the population proportion is 0.04 with a probability 0.90.

Answer: $n=404$

28. If $N=500$, $n=49$, $\sum_{i=1}^{49} X_i = 2940$ and $\alpha=0.05$, then determine confidence interval for μ . ($\sigma = 7$)

Answer: [58.138, 61.862]

29. A sample of size 64 units selected without replacement from a population of size 6000 units. If $\sum_{i=1}^{64} X_i = 3200$ and $\sum_{i=1}^{64} (X_i - \bar{X})^2 = 1575$, then estimate the population mean with 0.99 confidence level.

Answer: [48.4, 51.6]

30. According to the records of a hospital, heart disease was found in 52 of 1000 hospitalized men and 23 out of 1000 hospitalized women. Calculate the confidence interval for the difference of proportions with a 99% confidence interval.

Answer: [0.007, 0.051]